جامعة عفت

EFFAT UNIVERSITY

TravelPlanner

CS2071:Database system

Project

1 Introduction

TravelPlanner is a relational database system developed using SQL to manage travel bookings and related services. The system is designed to store and organize information about customers, flights, hotels, tours, transport, tour guides, and activities in one integrated database. The aim of the project is to support travel planning operations in a structured and efficient way.

2 Case Study

The idea of the project is to create a database system for a travel planner called **TravelPlanner**. The goal of the database is to track bookings for flights, hotels, and activities. The system stores information about customers, flight bookings, hotel reservations, tours, tour guides, transportation, and different types of activities that customers can book. :contentReference[oaicite:1]index=1

This project was selected because it contains many relationships that help in understanding relational database design and implementation. The database was created, designed, and developed using SQL. The report explains the case study, EER diagram, relational schema, table creation, data insertion, queries, views, triggers, joins, and delete operations. :contentReference[oaicite:2]index=2

3 Main Entities

The database includes the following main entities:

- Customer
- Flight
- Hotel
- Booking
- Tour
- Transport
- Tour Guide
- Activities

The system also includes relationship tables such as booking flights, booking hotels, assigning tours to customers, connecting tours with activities, and linking tours with tour guides and transport. :contentReference[oaicite:3]index=3

4 Database Description

The database stores customer information such as customer number, first name, last name, gender, age, email, and social security number. Flight data includes flight name, flight number, price, time, date, city/country, customer name, and description. Hotel data includes hotel name, room number, and room service. Tour data includes tour name and description. Tour guide data includes guide ID, guide name, and guide rate. Activities include restaurant names, tourist attractions, amusement parks, and garden parks. :contentReference[oaicite:4]index=4

The database also models several many-to-many relationships, such as:

- One customer can book many flights, and one flight can be booked by many customers.
- One customer can book many hotel rooms, and one room can be booked by many customers.
- One customer may have many transport tours, and one transport tour can belong to many customers.
- Tours can have many activities, and activities can belong to many tours.

5 EER Diagram

The EER diagram was created to represent the database entities, attributes, and relationships. It shows the structure of the system and the cardinality between entities such as Customer, Flight, Hotel, Tour, Tour Guide, Activities, and Transport. :contentReference[oaicite:5]index=5

Figure 1: EER Diagram of TravelPlanner

6 Relational Schema

After creating the EER model, the entities and relationships were converted into a relational schema. The schema includes base tables and relationship tables with primary keys and foreign keys to preserve the integrity of the database. :contentReference[oaicite:6]index=6

Examples of relation tables include:

- Booking_flight_customer
- Book_Customer_Hotel
- Tour_Trans
- Tour_have_TourGuide
- Take_tour_has_Activities
- Customer_may_have_Tour

Figure 2: Relational Schema of TravelPlanner

7 Implementation in MySQL / MariaDB

The database was implemented using MySQL / MariaDB. The first step was creating the database and selecting it for use. After that, all required tables were created with their attributes, primary keys, and foreign keys. :contentReference[oaicite:7]index=7

7.1 Creating the Database

The database was created with SQL commands and then selected for further implementation. :contentReference[oaicite:8]index=8

7.2 Creating Tables

Tables such as `Customer`, `Flight`, `Hotel`, `Book`, `Tour_Trans`, `Tour`, `Transport`, `Tour_Guide`, and relationship tables were created. Primary keys and foreign keys were used to maintain the relationships between tables. :contentReference[oaicite:9]index=9

Some examples:

- Customer table with customer number as primary key
- Flight table with composite primary key
- Booking table with customer and flight references
- Book table for hotel reservations
- Tour_Trans table for connecting tour and transport

8 Showing the Tables

After table creation, the database was checked using the `SHOW TABLES;` command. The system contains multiple tables including `activities`, `book`, `booking`, `customer`, `flight`, `hotel`, `may_havetour`, `tour`, `tour_guide`, `tour_have_guide`, `tour_trans`, and `transport`. :contentReference[oaicite:10]index=10

9 Inserting Data

Sample data was inserted into all major tables to test the system. This includes records for flights, customers, tours, transport, tour guides, hotels, bookings, and relationships between them. :contentReference[oaicite:11]index=11

Examples of inserted data:

- Flight records for airlines such as Etihad, Saudi Airlines, Qatar Airways, Emirates Airlines, and Flynas
- Customer records with names, ages, gender, email, and ESSN

- Tour records such as Tour One, Tour Two, and Tour Three
- Transport records connected to tours
- Hotel records with room numbers and services
- Booking records with total price and dates

10 Queries

A set of SQL queries was implemented to demonstrate retrieval, updating, filtering, and aggregation.

10.1 Retrieve All Hotels That Have a Spa

The system retrieves all hotels where the room service contains spa. :contentReference[oaicite:12]index=12

10.2 Update Flight Price

The price of flight number `EH1256` was updated to 530. :contentReference[oaicite:13]index=13

10.3 Count Male and Female Customers

A query was written to count the number of customers grouped by gender. :contentReference[oaicite:14]index=14

10.4 Retrieve Total Prices Greater Than 20000

The system retrieves booking records where the total price is greater than 20000. :contentReference[oaicite:15]index=15

10.5 Retrieve Customers With Names Starting With A

The system retrieves customer names that start with the letter A. :contentReference[oaicite:16]index=16

10.6 Average Flight Price

A query was used to calculate the average price of all flights. :contentReference[oaicite:17]index=17

10.7 Minimum and Maximum Booking Price

The system finds the minimum and maximum booking values grouped by customer. :contentReference[oaicite:18]index=18

11 Views

A view was created to display the full name and age of customers. The view concatenates first name and last name into one field and shows it together with age. This simplifies data retrieval and presentation. :contentReference[oaicite:19]index=19

12 Trigger

A trigger was implemented to update the `Guide_rate` column in the `tour_guide` table with the average rate of all tour guides whenever a new tour guide is added. This demonstrates the use of automation in the database system. :contentReference[oaicite:20]index=20

13 Join Operation

A join was performed between the `book` and `customer` tables to retrieve customer booking information. This operation demonstrates how data from different tables can be combined using relational principles. :contentReference[oaicite:21]index=21

14 Delete Operations

Delete operations were also tested in the system. Examples include:

- Deleting customers with age greater than 40
- Deleting customers with age greater than 33
- Deleting a specific customer using `Customer_No = 563431`

These operations show how records can be removed based on conditions or specific keys. :contentReference[oaicite:22]index=22

15 Results and Discussion

The project demonstrates how relational database concepts can be applied to a realistic travel-planning scenario. The use of EER modeling, relation schemas, primary and foreign keys, SQL DDL and DML, views, triggers, joins, and conditional queries helped build a complete and functional database system. The system supports structured storage and retrieval of travel data while preserving relationships among different entities.

16 Conclusion

TravelPlanner is a database project developed to manage travel-related bookings and services in a structured way. The project covers all major stages of relational database development, starting from case study analysis and EER design to relational schema, table creation, insertion, querying, view creation, trigger implementation, and deletion operations. It provides a useful example of how SQL can be used to design and implement a complete database system for real-world applications.

17 References

- Course project report and implementation screenshots
- EER diagram and relational schema files
- MySQL / MariaDB SQL commands used in the project